

Water Positive Intelligence: AI-Driven Monitoring & Efficiency

A major project report submitted in partial fulfilment of the requirement for
the award of degree of

Bachelor of Technology
in
Computer Science & Engineering

Submitted by

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Supervisor's Certificate

This is to certify that the major project report entitled '**Water Positive Intelligence: AI-Driven Monitoring & Efficiency**', submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is a Bonafide project work carried out under my supervision during the period from July 2025 to December 2025.

I have personally supervised the research work and confirm that it meets the standards required for submission. The project work has been conducted in accordance with ethical guidelines, and the matter embodied in the report has not been submitted elsewhere for the award of any other degree or diploma.

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Candidate's Declaration

We hereby declare that the work presented in this major project report entitled '**Water Positive Intelligence: AI-Driven Monitoring & Efficiency**', submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is an authentic record of our own work carried out during the period from July 2025 to December 2025 under the supervision of **Mr. Akshay Kumar**.

We further declare that the matter embodied in this report has not been submitted for the award of any other degree or diploma at any other university or institution.

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This is to certify that the above statement made by the candidates is true to the best of my knowledge.

Date: 29/11/2025

Place: JUIT, SOLAN

Supervisor Name & Sign: Mr. Akshay Kumar

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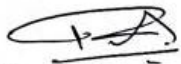
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Table of Contents

CHAPTER / SECTION	PAGE NO.
Supervisor's Certificate	i
Candidate's Declaration	ii
Acknowledgement	iii
Table Of Content	iv
List Of Tables	v
List of Figures	vi
List of Abbreviations, Symbols or Nomenclature	vii
Abstract	x
Chapter 1: Introduction	1
1.1 Introduction	
1.2 Problem Statement	
1.3 Objectives	
1.4 Significance and Motivation	
1.5 Organization of Report	
Chapter 2: Literature Survey	6
2.1 Overview of Relevant Literature	
2.2 Key Gaps Addressed by the Project	
Chapter 3: System Development	14
3.1 Requirements and Analysis	
3.2 Project Design and Architecture	
3.3 Data Preparation	
3.4 Implementation	
3.5 Key Challenges	
Chapter 4: Testing	23
4.1 Testing Strategy	
4.2 Key Testing Areas and Validation Metrics	
4.3 Test Cases and Outcomes	
Chapter 5: Results, Analysis, and Discussion	28

5.1 Presentation of Results	
5.2 Analysis of Outcomes	
Chapter 6: Conclusion and Future Scope	32
6.1 Conclusion	
6.2 Future Scope	
References	35

List of Tables

TABLE NO.	TITLE	PAGE NO.
Table 2.1	Literature Survey Summary	8

List of Figures

FIGURE NO.	TITLE	PAGE NO.
Fig 3.1	End-to-End Water Intelligence Architecture	17
Fig 3.4.1	Simplified Algorithmic Flow for Continuous Flow Leakage Detection	19
Fig 3.4.2	Operational Modules Diagram	21
Fig 4.1	Overview of the Testing Strategy	24
Fig 4.2	Key Testing Areas and Validation Metrics	27
Fig 5.1	Comparison Before and After Deployment	31

List of Abbreviations, Symbols or Nomenclature

ABBREVIATION / SYMBOL	FULL FORM
IoT	Internet of Things
KPI	Key Performance Indicator
CFLD	Continuous Flow Leakage Detection
CGWA	Central Ground Water Authority
BRSR	Business Responsibility and Sustainability Reporting
LPM	Liters per Minute
NFR	Non-Functional Requirement
FR	Functional Requirement
DB	Database
AI	Artificial Intelligence
ML	Machine Learning
ERP	Enterprise Resource Planning
API	Application Programming Interface
FIFO	First-In-First-Out Memory
LEED	Leadership in Energy and Environmental Design
SDG6	Sustainable Development Goal 6
Azure IoT Hub	Microsoft Cloud IoT Connectivity Service
MQTT	Message Queuing Telemetry Transport

Abstract

As a result of the scarcity of water, high cost of being involved with the water and environment, and regulatory mandate of the environment by agencies such as the EPA, especially in Industrial Manufacturing, Healthcare, and Utility industries the way water is being managed is changing drastically. There are many areas of Digital Data based applications in order to measure, analyse, and improve how water is used, however; when implemented in the real world, Industrial Water Management faces challenges such as poor or unreliable networking connectivity, the inability to provide consistent transmission of data, anomalies in data not being detected until after the fact, and not being able to extract useful data information from raw data streams coming from sensors.

This project represents the culmination of my internship at FluxGen Sustainable Technologies, one of India's leading companies focused on Industrial Water Management. My main focus is to improve upon the accuracy, reliability, and operational value of the Water Positive Intelligence System currently in use. My internship at FluxGen Sustainable Technologies has enabled me to conduct product research, evaluate hardware, assess client needs, and develop an in-depth understanding of the behaviour of industrial water consumption, which has culminated in the conceptualization and implementation of two significant innovations at the system level.

Contributions to the AquaGen data collection skid include:

1) Providing fault tolerant offline storage and synchronisation of AquaGen IoT devices' data. It does this with:

- A structured FIFO buffering system,
- Clear definitions of state changes, and
- A secure process for moving data from the IoT device and into the cloud.

2) Creating an artificial intelligence-assisted Continuous Flow Leakage Detection (CFLD) module that monitors consumption Behaviour to identify and alert users to abnormal usage while there is still time to act before damage occurs instead of waiting for large amounts of wasted product to accumulate, at which point corrective action will not be possible.

Some of the other deliverables that are part of this project are:

- (a) Enhancements to dashboard usability;
- (b) Creating a complete Product Requirements Document;

(c) Comparing the flow meters of the various manufacturers that compete against each other;
and

(d) Helping their customers become aware of their Water Balance, KPI's, and Consumption history.

The project will also provide enhancements to the Water Positive Intelligence platform to allow for more accurate data quality, a faster way of detecting anomalies, and more convenient accessibility to the platform for industries as they pursue their transition to becoming water neutral, while remaining within regulatory guidelines, and operating sustainably.

Chapter 1: Introduction

In today's 21st century, water has become one of the most valuable and sought after resources. Water is a critical component to the successful operation of commercial enterprises. Due to the serious lack of availability of drinking water in today's society, the shortage of such an essential resource is posing a direct and immediate peril to many companies and their ability to function. Companies that lack access to sufficient amounts of fresh water will be unable to attain their full production capacity and will also have to pay for compliance with governmental environmental guidelines and regulations.

Many companies today have an increasing concern about their future viability due to the increasing costs of obtaining fresh water supplies, as well as fulfilling customer demand while complying with the environmental requirements set forth by federal and state regulatory agencies such as the Central Ground Water Authority (CGWA), Business Responsibility and Sustainability Report (BRSR) enacted by Securities and Exchange Board of India (SEBI) and State Pollution Control Boards. As these entities continue to place pressure on formalized industrial regulations surrounding the use of water and have closed the door regarding the use of water as a "public good," industries will be required to establish new digital water ecosystems as a means of managing the inevitable global changes to the way water is used in business. These changing conditions require new systems that do much more than simply track the use of water. Today, businesses require continuous and real-time tracking of their use of water; this includes access to early warning signals indicating impending problems; predictive decision support information to assist with ecosystem planning; and automated documentation that can be delivered in audit-ready form on demand. Traditional methods and/or fragmented methods of tracking usage do not meet the rapidly evolving demands of the industrial sector.

FluxGen Sustainable Technologies is at the forefront of this evolution with the Water Intelligence Suite, which is an integrated platform containing durable hardware manufactured with IoT technology and communications devices manufactured with AquaGen wireless technology, along with the scalability of Microsoft Azure Cloud and intelligent analytics.

The system collects data from various parts of the industrial water cycle, such as flow, pressure, level, and quality, and provides them with useful insights using AI and machine learning. The project “Water Positive Intelligence/AI-Enabled Monitoring and Efficiency” is intent on enhancing this intelligence layer of the system. The intention of this project is to take large volumes of raw data from sensors which are supplied by the system and produce dashboards that show useful insights; alerts that allow users to act proactively based on the information contained in the dashboard, efficient operational workflows that make it possible to implement improvements with minimum disruption to current operations and documentation that supports sustainability efforts through automation. This intelligence layer of the system provides the clarity and precision needed to enhance decision-making capabilities of water-users as they move towards achieving their own water neutrality or, ideally, ultimately positivity.

1.2 Problem Statement

Continuous challenges with water resource management exist and have a negative impact on sustainability and operational performance. These same challenges create "blind spots" and prevent timely corrective action from being taken to address compliance concerns with respect to confidence.

- **Lack of and Inconsistent Data** – The majority of industries rely on manual meter readings and/or unreliable networks, which frequently fail due to intermittent failures. As a result of the occasional failure of these systems, gaps in the data created from these intermittent failures, result in no accurate way to analyse usage or create confidence in the use trend.
- **Undetected Leaks** – Continuous low flows caused by leaks going undetected during the night shift and other periods of inactivity often cause massive water waste and inflate utility costs; as a direct result, significant amounts of waste will add to inflated utility costs and create unreliable baseline consumption metrics.
- **Unable to Interpret Raw Data** – Industrial users do not typically receive processed data, such as LPH or pulse data, without context. Therefore, without intuitive visualisation and KPIs, it will be difficult for the management teams to turn unprocessed data into actionable steps. Disconnected

- **Operational Workflow** – It is common for operational workflows to operate independently from one another, such as when commissioning a system, verifying an issue with a customer, and reporting on sustainability efforts. The negative impact of having disjointed workflows is slow response times, poor communication within the organisational structure, and reduced trustworthiness of the monitoring system.
- **Regulatory and ESG Burden** – To comply with the CGWA guidelines, SEBI BRSR guidelines and Sustainable Development Auditing standard framework, an organisation must be able to accurately capture month to month tracking. Tracking these metrics manually can create a significant burden on the organisation in terms of administration and increased risk.

The Problem is to Create and Build an AI Powered Complete Capture of Water Insights into One Software Platform to Perform Error Detection/Leaking Detection with Great Accuracy and Improve Dashboard Interpretability, as Well as Automate the Operations Processes within an Organisation's Influence. The Ultimate Goal is to Assist all Industries in Achieving and Maintaining Water Positivity based on Fact-Based Evidence of What They Do with Their Water Resources.

1.3 Objectives

Despite being an overarching goal, The Industrial Water Use Project has three other areas to improve industrial water usage by altering the way we view the data used to make decisions regarding how we use industrial water:

1. **Make KPIs more visual** - Create Real Time Dashboards of KPIs showing Water Neutrality, Water Positive and Operational Efficiency of Industrial Facilities.
2. **Accurate Trend Analysis** - Develop effective methods for cleaning, interpolating and normalising data so that trends show accurate consumption patterns over daily, weekly, monthly timeframes.
3. **Anomaly Detection** - Develop a multi-tier anomaly detection system using both rule-based algorithms and Machine Learning models to identify anomalies including but not limited to, Continuous Leaks, Short Bursts (excess flow, pressure faults, Sensor Flaws).

4. **Automate Workflows** - Create automated end-to-end workflows from commissioning workflows through to Customer Issue Validation to reduce manual efforts and ensure repeated consistent execution.
5. **Create Automated Reporting** - Implement a system that automatically generates a monthly report on consumption trends, including detected leaky taps, estimated savings, etc., as well as action items for Plant Operators and Management.
6. **Help Address Sustainability Issues** - Create AI-Enabled Insights that assist industries with better practices including reduced waste, improve re-use and improve Water Consumption Patterns in alignment with Sustainable Development Goal #6 (Clean Water and Sanitation).

1.4 Significance and Motivation

- The significance of your project extends well beyond the technical. The project has a direct impact on the Resilience and Reputation of your industry. The shortage of water impacts not only production, but also the cost of operations and therefore affects compliance during audits. Any Company operating in an area that is vulnerable to drought, is under threat from water risk in the same way as business risk.
- Now, by transitioning from manual, reactive processes to a fully automated architecture (IoT + AI), companies will be able to:
 - Identify a Leak in a matter of hours instead of weeks
 - Reduce Waste in the operation by a significant amount
 - Base decision-making on factual data instead of assumptions
 - Address Compliance proactively and not reactively
 - Contribute toward Long Term ESG (Environmental, Social, Governance) Goals by Improving Sustainability Ratings.

The motivation for initiating your project is to fill the increasingly-necessary requirement of creating an effective set of transparent, measurable, and smart water management solutions that aid businesses navigate the rapidly changing Environmental and Regulatory Landscape. As businesses transition toward the use of Resources in a more Responsible Manner, there will be an increasing need for these tools to be utilized in an Effective Manner.

1.5 Organization of the Report

The progression and logical connection of the chapters in this report are as follows:

1. **Chapter 1** provides an overview of the project, detailed information related to the issues, the main objectives of the project, and gives background on why the system was created.
2. **Chapter 2** contains a complete review of the literature that includes information on Previously Conducted Research, Technological Frameworks, Existing Industrial Systems, IoT Architectures, and AI Methodologies for Modern Water Intelligence.
3. **Chapter 3** presents and explains how the system was developed starting with a requirements analysis, system architecture, data preparation pipelines, implementation decisions, and the key tools and algorithms utilized during the development process.
4. **Chapter 4** illustrates the testing methodology; how the system was validated, a description of all important test cases that were used, and the results of the tests performed both in simulated and real-world environments.
5. **Chapter 5** analyses the outcomes of the project, the system's performance evaluation (compared to other similar systems), the improvements that were made on the dashboard interface, the accuracy metrics of the various inputs and outputs of the system, and how the system was compared to conventional Water Management Systems.
6. **Chapter 6** concludes this report and presents opportunities for future development including the ability to integrate generative AI, automated decision-making systems, and predictive digital twin technology for industrial water ecosystems

Chapter 2: Literature Survey

2.1 Overview of Relevant Literature

To create the foundational technology base required for creating a Water Intelligence Ecosystem, an in-depth Literature Survey was undertaken through an analysis of fifteen Scholarly Articles, Industry White Papers, Technical Reports, and Standard Documents such as ISO, EPA, etc., published during the past five years. This assures that all the sources reviewed are relevant to today's Industrial Technologies and the governing bodies who make policies for Environmental Technology.

- The four major areas of interest related to Water Positive Intelligence are:
 - IoT and Data Acquisition
 - Cloud Infrastructure and Scalable Data Processing
 - Machine Learning & Analytics for (Water Networks).
 - Compliance with Sustainability Standards and Reporting Frameworks.
- Each publication provides different insights regarding IoT Sensors, Appendix D Systems, Regulatory Requirements, Artificial Intelligence Techniques used to predict consumption or identify illegal usage patterns (anomalous activity). The information included within these studies provides a great starting point to develop the next generation of Industrial Water Monitoring Solutions/Advisory Platforms.

I. IoT (Internet of Things) data acquisition

1. Rathore et Sharma (2020): They evaluated and compared LPWAN (LoRaWAN and NB-IoT) In large campuses, the impact of lost signals on battery life requires that intelligent scheduling occurs along with buffering to save data for retrieval later in the day.
2. Gurses et Ozkaya (2021): They stated that edge computing closes data retention gaps when communication is disconnected, but problems exist due to limited unit buffering depth and fault detection capabilities. Enhanced buffering will help solve these problems.
3. Li et Chen (2019): RS485, Modbus protocol is important for old technology plants; wiring up these machines can be a huge cost for many facilities. This led to the development of Hybrid solutions that utilize both wired and wireless connections.

4. Kaur et Singh (2022): Rainwater and Used Water Management Frameworks can help provide a basis for rain and used water Integrated Water Resource (IWR)/Used Water Integrated Water Resource management modules. Also, to optimally utilize these modules calibration must be specific to the intended use (i.e., location).

II. Cloud Architecture and Data Processing

5. Microsoft (2023): Azure IoT Hub and Cloud Native Tools enable the ingestion of telemetry data at scale. The design of the Azure IoT Hub is Cloud Agnostic which mitigates the potential vendor lock in risk to their clients.
6. Wang et Zhou (2021): Verification of real time “Anomaly” detection using low-latency streaming processing technologies was achieved; however, due to the operational cost factor batching and optimizing would have to occur to keep the costs manageable.
7. Al-Ghamdi (2020): The quality of sensor data increases with the usage of data smoothing and interpolation techniques, and parallel techniques for ruling out hidden anomalies will continue to be developed.

III. Machine Learning Techniques in Water Systems

8. Mounce and Boxall (2018) – Better accuracy of leak detection using supervised learning; integrated with rule-based systems, reduces the amount of labelled data required by the supervised learning model.
9. Liu and Zhou (2019) - The use of an Isolation Forest algorithm allowed for the identification of rare anomalies, while the addition of multiple filtering layers helps to reduce false positives.
10. Zhang and Li (2022) - Long Short-Term Memory Networks (LSTMs) are capable of accurately predicting long-term water usage, and through the optimization of model architecture can significantly reduce the need for computational resources.
11. Sisinni and Leali (2023) - The use of Digital Twin technology allows us to create virtual models of water systems, requiring significant resources to do so, while providing great potential benefit in terms of the ability to perform predictive maintenance and run simulations.

IV. Reporting on Compliance and Sustainability

12. SEBI (2021) - Water-related disclosure requirements were outlined in the BRSR and automated advisory modules were developed to deploy operational guidance.
13. The implementation of new techniques for extracting groundwater, measuring groundwater quantity and quality, developing KPIs to measure the extraction and monitoring of groundwater (quantitative and qualitative), and providing a flexible way to measure these key indicators on an ongoing basis due to changing groundwater regulations, was initiated by CGWA.
14. The United Nations Global Compact established Sustainable Development Goal Number Six (SDG 6) as a reference point to assist in developing Action Plans for the Sustainable Use of Water and to bridge the implementation gap related to this important area of Sustainable Use of Water.
15. The identification of water and energy waste and the development of Intelligent Monitoring techniques (Siddique & Alam) provide ways to decrease operating costs and increase resiliency.

Table 2.1 Literature Survey Summary

S.NO.	AUTHOR(S)	YEAR	FOCUS AREA	KEY FINDINGS (PROJECT RELEVANCE)	LIMITATIONS (PROJECT ADDRESSES)
[1]	Rathore, J. & Sharma, A.	2020	LPWAN Protocols & Sensors	Validates LoRaWAN/NB-IoT for range and low power; informs AquaGen design.	High cost, poor signal penetration, and limited battery life at high frequency.
[2]	Gürses, B. & Ozkaya, H.	2021	Data Reliability & Edge	Necessity of Edge Computing	Buffer capacity limits outage tolerance; does

				(buffering) to prevent data loss during network outages (NFR-2).	not detect sensor physical failure.
[3]	Li, S. & Chen, X.	2019	Industrial Protocols	Requirement to support wired standards like RS485 for hardware compatibility.	Wired connections increase installation and maintenance costs on large sites.
[4]	Kaur, M. & Singh, J.	2022	Specialized Water Modules	Framework for Rainwater/Used Water Intelligence (RWI/UWI) module development.	Optimization modeling requires complex, context-specific calibration for accuracy.
[5]	Microsoft Corp.	2023	Cloud Scalability	Confirms Azure IoT Hub and Time-Series DB use for highly scalable data ingestion (NFR-1).	Risk of vendor lock-in and demands high cloud management expertise.
[6]	Wang, H. & Zhou, L.	2021	Big Data Stream Processing	Architecture for low-latency processing via Event Hubs/Stream	High operational expenditure (OpEx) for continuous real-

				Analytics (NFR-4).	time data streaming.
[7]	Al-Ghamdi, H.	2020	Data Quality Management	Recommends interpolation (e.g., STL Decomposition) for handling data gaps and continuity.	Interpolated data can potentially mask underlying short-term sensor anomalies.
[8]	Mounce, S. & Boxall, J.	2018	Anomaly Detection (Supervised)	Confirms high accuracy of supervised classifiers for identifying labeled leakage events.	Requires extensive, accurately labeled historical data for effective training.
[9]	Liu, F.T. & Zhou, Z.H.	2019	Anomaly Detection (Unsupervised)	Methodology using Isolation Forest for detecting novel, unseen consumption anomalies (CFLD enhancement).	Unsupervised models often result in a high False Positive Rate (FPR).
[10]	Zhang, Y. & Li, J.	2022	Consumption Forecasting	Effectiveness of time-series models (e.g., LSTM) for creating accurate	LSTM models are computationally expensive and demand significant

				baseline consumption models.	training resources.
[11]	Sisinni, E. & Leali, F.	2023	Digital Twins	Conceptual foundation for a Digital Twin to enable Predictive Maintenance (Future Scope).	Development is resource-intensive and requires integration of multimodal sensor data.
[12]	SEBI (India)	2021	ESG Reporting Standards	Defines mandatory metrics and format for the BRSR report (FR-5).	The framework is only a reporting standard, lacking operational guidance.
[13]	CGWA (India)	2020	Regulatory Compliance	Defines metrics for Groundwater Withdrawal and Water Neutrality KPI calculation.	CGWA regulations are subject to frequent regional amendments and variations.
[14]	UN Global Compact	2019	Global Sustainability Metrics	Provides context for the goal of achieving and verifying Water Positivity (aligned with SDG6).	SDG targets are aspirational and lack immediate, detailed, and legally binding audit requirements.

[15]	Siddique, A. & Alam, R.	2018	Industrial Water-Energy Nexus	Details the financial and operational risks of poor water use efficiency in industry.	High-level strategic review that lacks prescriptive engineering solutions for implementation.
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2.2 Key Gaps Addressed by the Project

The following are the significant gaps the project intends to fill as a result of this Project's research.

- The solution addresses the problem of fragmented platform ecosystems where traditional smart water systems continue to operate in silos by combining hardware-agnostic IoT with cloud-based system/service provider processing (via Microsoft Azure) and AI (artificial intelligence) analytics pre-configured for the specific needs of the market in one integrated solution that enables continuous monitoring and control of the entire system.
- There is currently a limited ability to detect subtle leaks. Most available systems today use low-flow thresholds to try and identify a leak and miss most of the nights or low usages. The Continuous Flow Leakage Detection (CFLD) engine identifies very subtle anomalies that a conventional low-flow or night usage threshold would miss.
- Historically, there has been a lack of connection between the sensor data generated by older-style systems to an operational or managerial decision. This project uses automated advisory options to allow users to take advantage of the analytics generated by the CFLD engine and streamlines their processes from commissioning to review.
- Traditionally, report generation has followed a retroactive method of describing water consumption trends rather than collecting and reporting data in real-time. This project is providing users with a dynamic set of Water Positivity KPIs and dashboards for managing their

water-use sustainably now, as well as going into the future based on able to comply with local regulatory requirements (CGWA and SEBI BRSR).

Existing legacy systems do not have the capability or means to accommodate the amount of high-frequency sensor data produced by several sensors, nor can they adequately scale to do so for an industrial-scale deployment. Therefore, the platform was developed to handle large amounts of data via the use of cloud processing and optimised analytic machinery that allow for efficient management and scalability of operation.

Chapter 3: System Development

In this chapter, I detailed the methodology used to design and build the Water Positive Intelligence system. I first performed an in-depth analysis of the requirements of our industrial clients, as well as the challenges associated with traditional methods of managing water to determine the need for a data-driven predictive approach. I then described the complete architecture of the Water Positive Intelligence system, including all of its components, such as the edge layer, ingestion to the cloud, processing engine, storage, and application layers, as well as how these various components interact to provide real-time water intelligence in an easy-to-use way.

After discussing the architecture of the system, I will go into detail about how to prepare data to be used for artificial intelligence and analytics models when they operate on accurate and complete data. Specifically I will describe the techniques used to prepare the data for accurate AI and analytics model operation by discussing cleaning, validating, interpolating, and extracting features from the data for final preparation of the data to be used to create a complete dataset for our core AI module for leakage and anomaly detection, which is based on rule-based logic combined with advanced machine learning models to identify continuous-flow leakage and other irregular consumption of water.

The final chapter illustrates the operational modules of the Water Positive Intelligence system by providing a summary of each operational module, a clarification of how each module will assist stakeholders in measuring, assessing, tracking and optimizing their water use in compliance with applicable laws and regulations, and a description of significant hardships experienced during the creation of the modules (e.g., data gaps, false positives, integration with legacy SCADA systems, user error/misinterpretation and environmental noise) along with how we resolved those difficulties.

Therefore, this chapter represents a complete and comprehensive overview of the evolution of the Water Positive Intelligence system from beginning to end and provides a viable and dynamic solution for managing industrial water usage.

3.1 Requirements and Analysis

Water Positive Intelligence offers industries a transition from inefficient, reactive water management methods to a more predictive, proactive, and data driven ecosystem. Prior to

starting the development, it is essential to identify both functional and non-functional requirements before beginning development. This is to ensure that the system meets both the expectations of clients as well as the industry standards.

Functional Requirements (FR)

1. **FR-1 Sensor data acquisition (real-time)** – The system will serve as an active collector of the data coming from field sensors. This information includes measurement of flow, pressure, water level and rain fall. By acquiring this information in real-time, timely detection of issues with a quick response time for necessary decisions, can be made.
2. **FR-2 Continuous Flow Leakage Detection (CFLD)** – Many times industrial leaks go undetected because they happen while the plant is non-operational. This feature provides the ability for detection of leaks that are occurring at low flows continuously therefore providing the end-users the ability to act quickly to mitigate the losses and preserve the water resources.
3. **FR-3** A visual representation of plant wide water balance and usage will be provided in the form of a dashboard, real-time and available to all stakeholders to monitor the health of the system.
4. **FR-4** The system will automatically generate monthly reports that will summarize key performance indicators (KPI) and all anomalies, trends in water consumption etc, along with prescriptive recommendations for improvement.
5. **FR-5** Both WII and WNS provide quantitative measurements to evaluate a facility's water efficiency/use and sustainability. By having the WII and WNS as measures of the plants performance relative to the overall goal of being 'water positive' (WPT), they will allow you to assess the facilities ability to operate efficiently with water resources.
6. **FR-6:** Integration with SCADA, Geospatial, and Manual Inputs The system should provide one view of all of the data by integrating all relevant data from existing SCADA systems, of external sources (geospatial) and records of monitored data entered by operator(s).

Non Function Requirements (NFR)

1. **NFR-1: High Scalability** - The database must have the ability to maintain many millions of data points per day being supplied by multiple plants and is expected to grow under high loads.

2. NFR-2: Buffering (Fault Tolerance) - Edge devices must store information through First In First Out (FIFO) Cache Memory until connections are restored and will not lose any data due to disconnections at Edge devices.
3. NFR-3: Secure communication - Data transferred from the Sensor to the Cloud must be encrypted by one of the secure methods of transferring data for IoT devices (i.e. secure protocols such as MQTT and HTTPS).
4. NFR-4: Dashboard (Update less than 1 second) - Dashboards will need to be updated with information in near real time to provide operators with the ability to act to an event as fast as they can.
5. NFR-5: Modular Cloud Architecture – Server-less / Microservice Architecture allows for maintainability, flexibility, and scalability when developing future enhancements.

3.2 Project Design and Architecture

The Project has designed a modern IoT system architecture and the modular nature of the Project enables the Project's devices and cloud services (AI modules) to easily integrate with one another. The Project's modern IoT architecture provides a high degree of flexibility, scalability, and the ability to perform real-time data analysis.

1. Edge Layer – AquaGen Device

- a) Gathers data from sensor measurements of flow rate, pressure, levels, and water quality.
- b) Acts as a temporary buffer for the data using first in-first out (FIFO) memory in case of short-term disconnection from the cloud.
- c) All data is encrypted prior to transmission to the cloud for secure transmission.

2. Cloud Ingestion Layer - Azure IoT Hub

- a) Manages the simultaneous ingestion of millions of data points.
- b) Manages authentication and identity of devices and provides security to the entire IoT system.
- c) Facilitates the routing of telemetry data to downstream cloud-based services to be stored, analyzed and visualized.

3. Processing Layer – Analytics Engine

- a) Cleans the data by removing noisy data and creating baselines for normal operation.
- b) Calculates key performance indicators (KPI's) related to usage of water, unaccounted losses of water, and potential leaks.
- c) Executes Artificial Intelligence (AI) & Machine Learning (ML) algorithms to perform anomaly detection, predictive modeling, and optimization of system performance.

4. Storage Layer – Time-Series Database

- a) Saves very high-frequency and time-stamped sensor records for later analysis or longitudinal tracking.
- b) Provides fast response time for dashboard, report and training purposes.
- c) Guarantees that data will be reliable and available for trend analysis and evaluation of performance.

5. Application Layer - Reporting & Dashboarding

- a) Real-time water consumption/usage, consumption/usage trends, and alerts to the stakeholders of an organisation are displayed on a dashboard.
- b) A dashboard can also show the ground water recharge + ground water balance and key performance indicators.
- c) Also includes monthly and custom reporting options to assist in decision-making and meeting regulatory requirements.

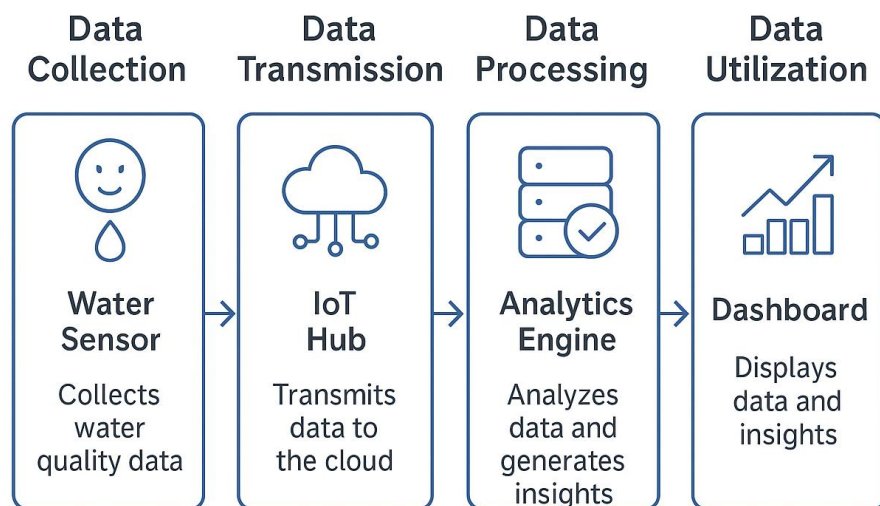


Figure 3.1 – End-to-End Water Intelligence Architecture

3.3 Data Preparation

Preparing data through proper cleaning and validation to remove irrelevant data that can result in inaccurate AI predictions, KPI Computation. Additionally, prepared data is consistent in many different areas.

1. **Cleaning & Validation of Data** - Removes electrical noise, spikes, invalid readings (negative flow or out of range).
2. **Interpolating Data** - Shifts the data forward by creating link holes in the data using linear interpolation to connect short gaps of time-series data (including trends).
3. **Establishing Baseline Points** - For each meter: A normal consumption trend for both operational and non-operational hours. Baseline points serve as anomalies when predicting.
4. **Generating Features and Extracting KPIs** - Aggregate low-performance meters into higher-level metrics (Water Intensity Index and Water Neutrality Status) so the KPI provides meaningful data for dashboards and reporting purposes.

3.4 Implementation

3.4.1 Leakage & Anomaly Detection Engine

Implementation of Leakage and Anomaly Detection

Our approach to leak and anomaly detection consists of two tiers:

- A Rule-Based Tier detects simple static leak conditions, e.g., tank level drops below a set point or a critical failure detected by Rule-Based Tier.
- The CFLD (Continuous Low-Flow Detector) Tier alerts system operators when a sustained change occurs and provides information on actual, critical leak conditions.

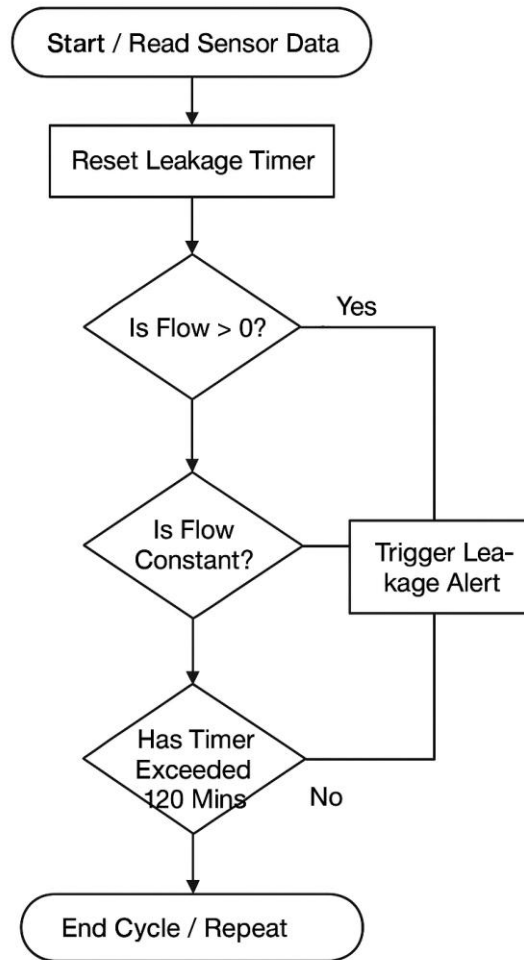


Figure 3.4.1 Simplified Algorithmic Flow for Continuous Flow Leakage Detection

3.4.2 Operational Modules

Module: Business Operation Tracking

- Provides Real-Time Data to Employees, Supervisors, and Sustainable Development Groups
- Includes Current Water Levels and Historical Water Use Over Time
- Provides the Following Overview of the Aquifer Recharge Patterns (Recharge vs. Consumption) Over Time: Recharge Patterns of Increased or Decreased Aquifer Recharge, and Status of Meter Usage
- Alerting Employees to Potential Leaks or Other Anomalies; High-Risk Situations
- Employees Will Be Able to Verify (a) If Their Meter is Functioning and (b) if Their Meter is Connected to the Network

Monthly Report Generation:

- Automatically create metrics for each month related to water usage and the performance of the system;
- Summary reports of all key metrics (e.g. total water consumed, water that has been lost, aquifer recharge, tanking activity during peak water consumption);
- Store records of all the different water-related anomalies, including leaks and deviations from the normal operational limits;
- Provide instructions for companies on how to increase their overall operating efficiencies as well as a list of actions that need to be taken;
- Provide the ability to monitor the progress of becoming "water neutral" and find opportunities for savings;

Commissioning to Review:

- Help customers go through the initial start-up of their facility as part of the onboarding process.
- Validate the accuracy of the sensors and data collected from the sensors during the commissioning process.
- Have the Auxiliary Engineer review and document that the manufacturing facility is working within the baseline established during the first 30 days of the project.
- Make sure that all of the pipelines, sensors, pumps, and other equipment are documented and set to the proper position on the mapping.
- Support the standardization of the process for reviewing and documenting water-related projects by the industry.

3.5 Primary Challenges

Data Gaps and Connectivity Failure: If a network fails or has a weak signal, AquaGen's FIFO storage system will temporarily hold all the data until the connection is re-established. However, any brief connectivity gaps will still be bridged by interpolation so data continuity is maintained.

- **Missed Alert Signals:** Any sudden spike in activity may cause a false alarm. The use of historical data along with a 30-minute sustained deviation will allow for the reduction of false alarm opportunities.
- **Legacy SCADA Integration Issues:** Some legacy SCADA systems use different and incompatible protocols. In order to provide a streamlined integration of IoT modules with SCADA system, Protocol Converters and Application Programming Interfaces (APIs) will need to be created to provide standardized data flows.
- **Operator Misinterpretation of Dashboard Information:** An operator could easily misinterpret dashboard data and/or key performance index (KPI) information. The use of more intuitive dashboarding, additional plain-language artificial intelligence (AI) alerts that advise users, will allow for improved operator understanding of dashboard information for decision-making purposes.
- **Environmental Noise Interference and Sensor Reading Inconsistency:** Certain industrial conditions can produce misleading or erratic sensor readings. Noise filters and smoothing algorithms used in addition to regular sensor calibrations can help maintain accurate data representations of sensor readings.

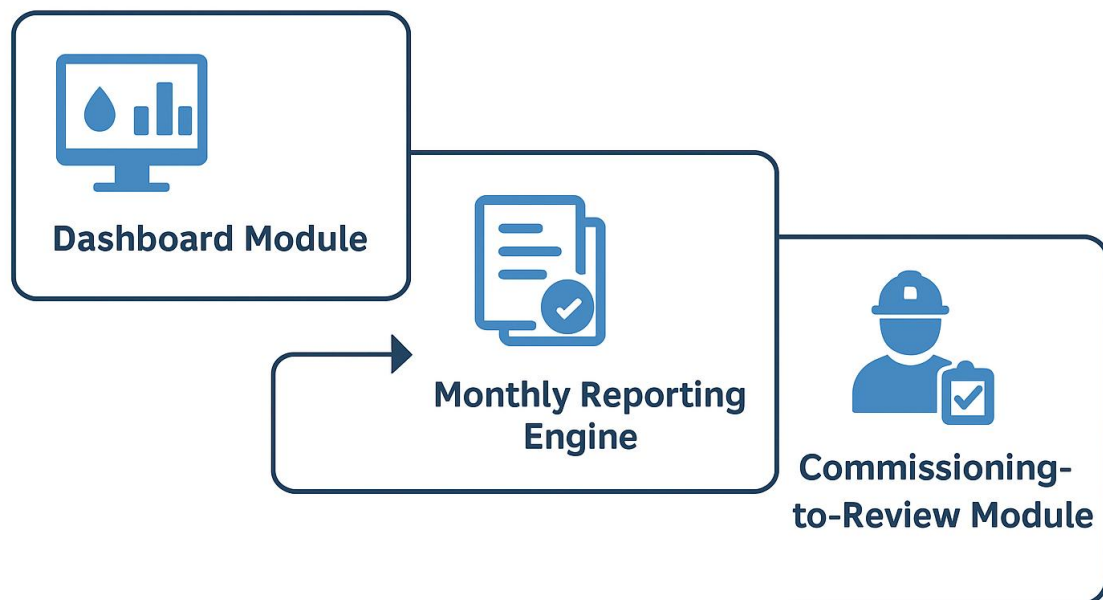


Figure 3.4.2 Operational Modules Diagram

3.5 Key Challenges

- **Data Gaps & Connectivity Loss:** Networks may fail or have poor signal. AquaGen's FIFO buffer stores data until the connection is restored. Interpolation fills short gaps to maintain continuity.
- **False Positives in Alerts:** Sudden spikes can trigger false alarms. Using historical data and a 30-minute sustained deviation rule reduces these errors.
- **Legacy SCADA Integration:** Older SCADA systems use incompatible protocols. Protocol converters and APIs standardize data flow for smooth integration with IoT modules.
- **User Misinterpretation:** Operators may misread dashboards or KPIs. Intuitive visuals and plain-language AI advisories improve understanding and decision-making.
- **Environmental Noise & Sensor Inconsistencies:** Industrial conditions can cause irregular readings. Noise filtering, smoothing algorithms, and regular sensor calibration ensure data accuracy.

Chapter 4: Testing

This chapter outlines the details of the testing method that was employed to confirm that the Water Positive Intelligence product functions as expected. Testing provides confidence in the product's operation; that its anomaly detection function works; that it meets both functional requirements (FR) and non-functional requirements (NFRs), and that it performs well across various types of industrial environments. Because the Water Positive Intelligence product uses multiple components (hardware components in the form of sensors and software in the form of cloud analytics) along with AI algorithms to make operational decisions, a thorough and layered testing method was used to validate the results at each point within the data pipeline.

The overall goal of testing was to validate and confirm that the Water Positive Intelligence product produced accurate results, that it is dependable, and that it will work for potential end-users in their real-life applications of the product, including; situations in which a network outage occurs, the use of large amounts of data to detect less frequent events, and unpredictable operations. In addition to detecting and identifying any faults in the system, testing enables the use of objective, measurable data regarding the product's ability to meet these requirements.

4.1 Testing Strategy

Testing for the Water Positive Intelligence system is executed at various levels to ensure that hardware, as well as the software components of this system, are fully integrated into the testing process and fulfill their individual purposes.

The purpose of the testing strategy was to verify that the collected sensor data is being recorded correctly, AI algorithms will generate the appropriate alerts and that the dashboards and reports generated by the system will provide the customer with actionable information.

The key objectives of the testing strategy included:

1. Sensor Data Integrity – (NFR-2)

Testing of edge devices (AquaGen) was conducted to ensure that they reliably capture sensor readings regardless of network outages and that they buffer and store readings during the times when they do not have a connection to the network. Loss of data can create problems for both the AI algorithms and calculating KPI's. The test environment was designed to simulate actual

field conditions to verify the effectiveness of the FIFO buffer. In addition, testing confirmed the ability of the system to download back to the edge device once connection is re-established.

2. Functional Testing

Functional requirements were tested and met. This functionality includes the real-time collection of sensor data, the ability to generate an automated monthly report, the ability to generate a Water Balance Calculation, the Water Intensity Index, and the Water Neutrality Score. Functional testing for normal and stress conditions confirmed that all functional requirements were met and that the system performed all required functions correctly.

We'll look at the accuracy of alerts Created with AI as well as Continuous Flow Leakage Detection (CFLD) being a major component of system "A" therefore was tested with various simulation scenarios in order to ensure appropriate and timely alerts are produced during non-operation times by evaluating through historical and simulated datasets for precision and recall as well as latency of producing alerts to confirm operators receive reliable timely alerts without generating any false alerting information.

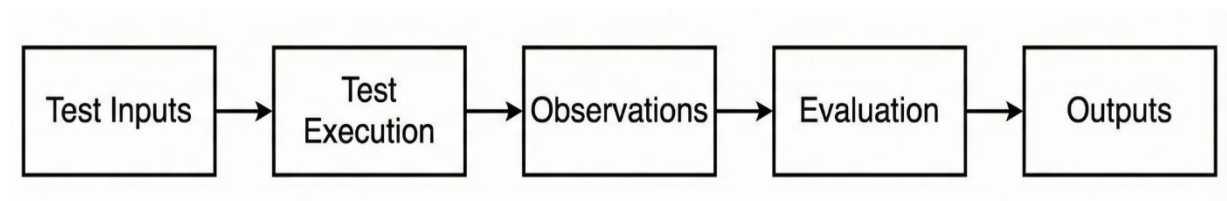


Figure 4.1: Overview Of the Testing Strategy

4.2 Key Testing Areas and Validation Metrics

4.2.1 Sensor Data Integrity and Fault Tolerance

The testing for this area concentrates on testing the ability of edge devices to remain operational, storing/forwarding messages to their destination (cloud database) without interruption due to network outages.

- **Testing Method:**

Testing was accomplished by intentionally removing all network connectivity for a total 14 hour period (exceeding 12 hour NFR threshold). While connectivity was lost sensors continued recording measurements (water flow, pressure and level) locally (FIFO buffer) until network

connectivity was restored and User-Defined Procedures (UDP) would take over and synchronize the buffered measurements to the cloud database.

• **Validation Metrics:**

DLR is the percentage of all sensor readings that didn't make it to the cloud database while the sensor was offline. A 0% DLR means that all readings were successfully returned to the cloud, and in the order they were taken.

Accuracy of Data: Ascertained that measurement values were accurately transmitted back to the user and that the corresponding timestamp was accurately tracked in the user's record. Successful testing of this area will demonstrate the ability of the system to maintain accurate and dependable data over time while under extreme conditions in the field.

4.2.2 AI Alert Accuracy Testing

The CFLD algorithm has been evaluated using precision, recall, and latency metrics, which are essential to demonstrate that an AI solution can perform as intended.

- Method: The AI module was validated using both
- Training Data: Historical labeled events indicating a known leakage.
- Blind Test Data: Simulated operational patterns related to normal consumption; rapid surge(s), and sustained night-time flow(s) leakage.

Metrics Used for Validation:

- Precision: Proportion of alerts corresponding to actual leak(s). High precision will minimize the number of false positive alerts and will reduce the number of unnecessary operations by operators.
- Recall: Proportion of actual leaks detected and identified. A high recall rate will ensure that there are no critical leak(s) missed, thus ensuring the protection of our water resources and the mitigation of water loss.
- Alert Latency: Time from sustained leak(s) detected at the Cloud Ingestion Point to Alert Display on a Dashboard. The shorter the alert latency, the quicker action can be taken by the operator.

- The results of the testing indicate that the AI system will accurately identify subtle anomalies in addition to not having excessive false alerts.

4.2.3 Performance and Scalability Testing

Performance testing validates the System's ability to deliver high-volume throughput and low latency.

- **Test Procedures:** For this performance test, the Azure Load Testing Service was utilized to simulate a gradual increase in sensor data stream load, and a test was conducted to evaluate how the IoT Hub, Analytics Engine and Time Series Database performed under the maximum threshold of 10,000+ message per second. Performance was tracked to see if the System could provide real-time and accurate service to users and how long the Analytics Engine took to provide that service to users.
- **Validation Metrics:** The number of messages that were successfully processed every second.
 - o **Latency of Dashboards:** The amount of time that elapsed between a dashboard requesting to display data from the database and when the data was displayed on the dashboard. The System successfully delivered user requests in a latency of less than one second at its peak load which validates the System has the capability to provide responsive service and to be, therefore, scalable for deployment to industrial organizations.

4.3 Test Cases and Outcomes

The following table presents the sample set of high priority test cases that were used to validate the functional performance and basic operational functions of the Water Positive Intelligence system.

The results of these tests confirm that the Water Positive Intelligence is a scalable and reliable product that meets all functional and non-functional requirements. The test results of the Water Positive Intelligence system included the testing of sensors, AI testing, performance testing, and the usability of dashboards to ensure that the product can be successfully deployed into real production environments.

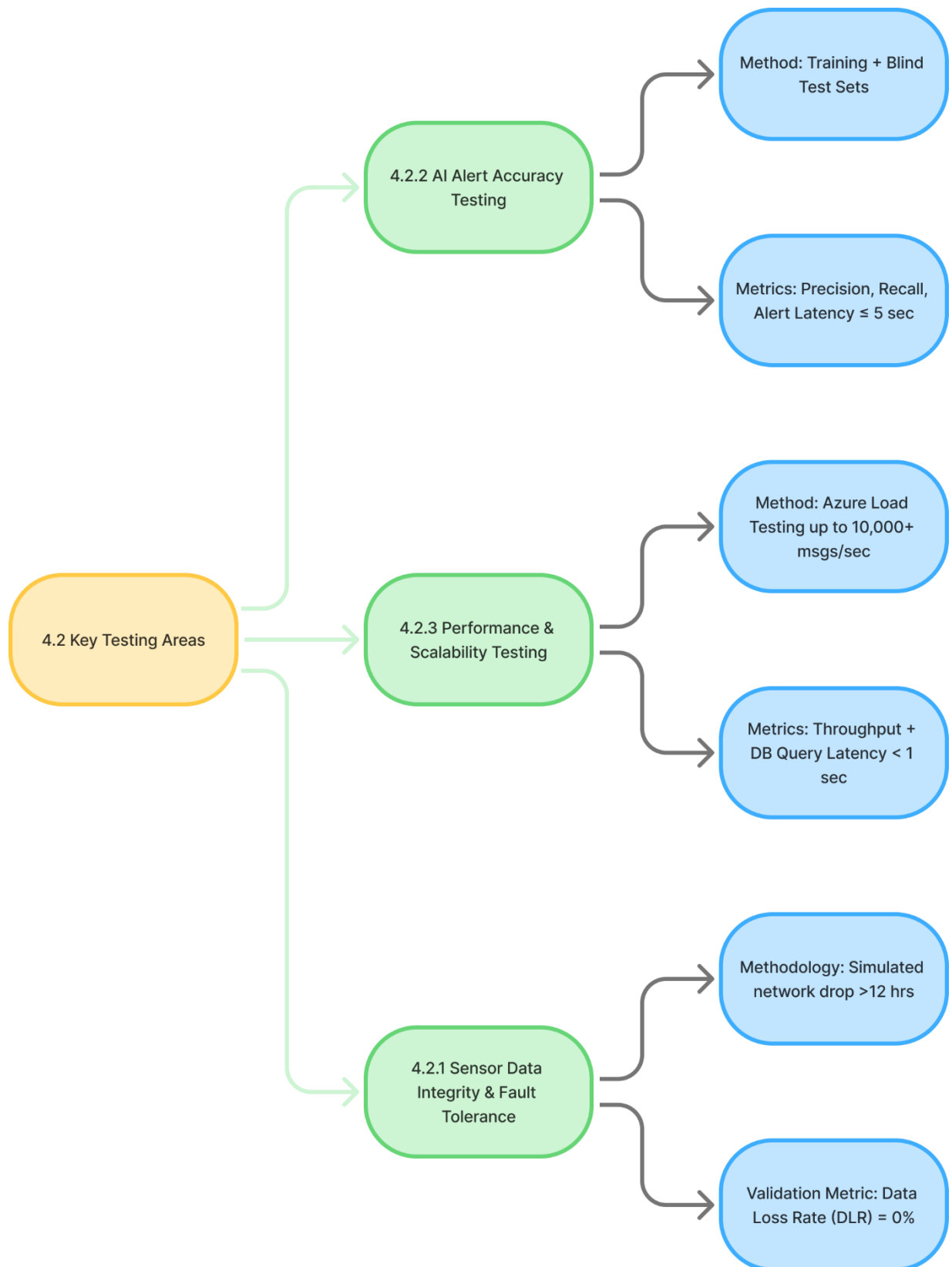


Fig 4.2 Key Testing Areas and Validation Metrics

Chapter 5: Results, Analysis, and Discussion

The results of the Water Positive Intelligence deployment are included in this chapter. The analysis details how to improve the performance of industrial water management, as well as the implications for the future of this industry as we know it today. The data gathered on both the quantitative side of the project contains information about the savings achieved by using the Water Positive Intelligence system, the prevention of future leaks, and the financial losses incurred through leakage(s). The qualitative data gathered during the project shows that increasing operational efficiency, improving compliance and readiness for future compliance audits, and being able to report on sustainability offers value to a business and demonstrates how they can be successful in their water management program.

5.1 Results Presentation

This section focuses on showing the actual performance at various client sites and how those performances lead to measurable improvements in operating efficiency (savings), financial (monetary) savings, and sustainability improvement (environmental).

Quantitative Results

The results listed below reflect measurable and verifiable improvements resulting from implementing the Water Positive Intelligence platform:

- **Water Consumption Decrease:** A demonstrable reduction in total water consumed was achieved by multiple client sites. For instance, Lactalis had a 19% decrease in total water used over 10 months which is a meaningful reduction in total daily water consumption. The ability to continuously monitor water used and be aware of real-time notifications regarding leak detection, flow anomalies, and inefficiencies provided the opportunity to see these problems before they could negatively impact their operations.
- **Financial Savings:** The decrease in total water consumption also translates into financial benefits to those companies as well. Savings experienced by Lactalis broke down to approximately INR 17.5 lakh/year, which illustrates how offering real-time monitoring and detection of water anomalies can ultimately generate real money savings for

organizations. Such monetary savings will ultimately support the business case for expanding the Water Positive Intelligence solution into more production facilities.

- **Leakage Metrics** - The CFLD algorithm was developed to assist in finding ongoing flow leakages that go undetected for long periods. It allows for the calculation of the total kilolitres of water saved after completing leak mitigation, and it also shows the CFLD's ability to detect both hidden and slow flow leaks, which collectively create significant amounts of water being lost.
- **Water Positivity Status** - The system has enabled clients to assess their progress toward meeting their regulatory sustainability goals, including Water Neutrality and Water Positivity. By providing ongoing monitoring and recommendations on corrective actions, many sites have been able to document their progress toward achieving both the National and World Water Use Targets, with several sites accomplishing at least Water Neutrality, if not achieving Water Positivity in most cases.

Qualitative Results

While there are quantifiable benefits, several qualitative benefits demonstrate the wider positive effects of the programs:

- **Regulatory & Sustainability Accomplishments:** Environmental compliance scores significantly improved for many client locations as well as the potential for many client locations to achieve certification(s) for example, using efficiency enhancements from water conservation allowed these facilities to obtain LEED Platinum certifications, which are the premier Green Building rating system and have the highest standard for environmental-based certifications with strict criteria for achieving them.

Operational Effects: Due to the ability to send out a message/alert in real time, and also provide on-going monitoring and diagnostics of potential failure modes, this resulted in fewer emergency service calls for plant employees as they were able to rely on the diagnostic information provided by the system to assist them in planning future maintenance activities, making it possible to have a shift toward more proactive maintenance practices, providing less frequency of unscheduled plant outages and improved overall operational reliability.

5.2 Analysis of Outcomes

The results of this section will provide an insight into how the improvements were achieved and how the technical and architectural decisions made in Chapter 3 contributed to those results.

- The AI Module was effective in detecting leaks because of its ability to utilize a machine learning algorithm to detect leaks and its ability to identify patterns of water usage, especially during low-flow hours, enabling it to identify slow, long-term leaks that would otherwise go undetected by traditional systems. AI alerts were found to accurately corresponded with actual leaks being present at the location where the alerts were generated, thus establishing the AI's accuracy and reliability in detecting leaks. This evidence supports the use of both a machine learning and a rule-based approach to the development of the AI algorithm's detection logic.
- KPI's Role and Effect - As a result of implementing real-time dashboards (or KPI views) and specifically, the Water Balance view and Water Neutrality score, plant operators are now able to see exactly how much water they are consuming at each point of the operation; in addition, users are able to monitor their consumption levels as they change throughout the day.
- With constant access to real-time consumption data and how it varies in response to changing usage patterns, plant operators are now using the information provided to them to better modify their operating schedules based on usage trends; perform increased inspections of pipelines based on historical data collected and responding more rapidly to system alerts than in the past.
- The fact that plant operators are relying on data more, than ever before, for making operational decisions has helped lead to increased conservation of water use as well as increasing the operational efficiency of the facilities they operate.
- Testing and Verification of Infrastructure Efficiency - In addition to being able to view the consumption data and respond more effectively, the cloud-based infrastructure being deployed using Microsoft Azure has already shown to perform exceptionally well in a live production environment. Specifically, it has been shown to adequately support sensors that produce extremely high frequency data and provide alerts using very low latency and gives users the ability to view sensor data in real-time through the dashboards.

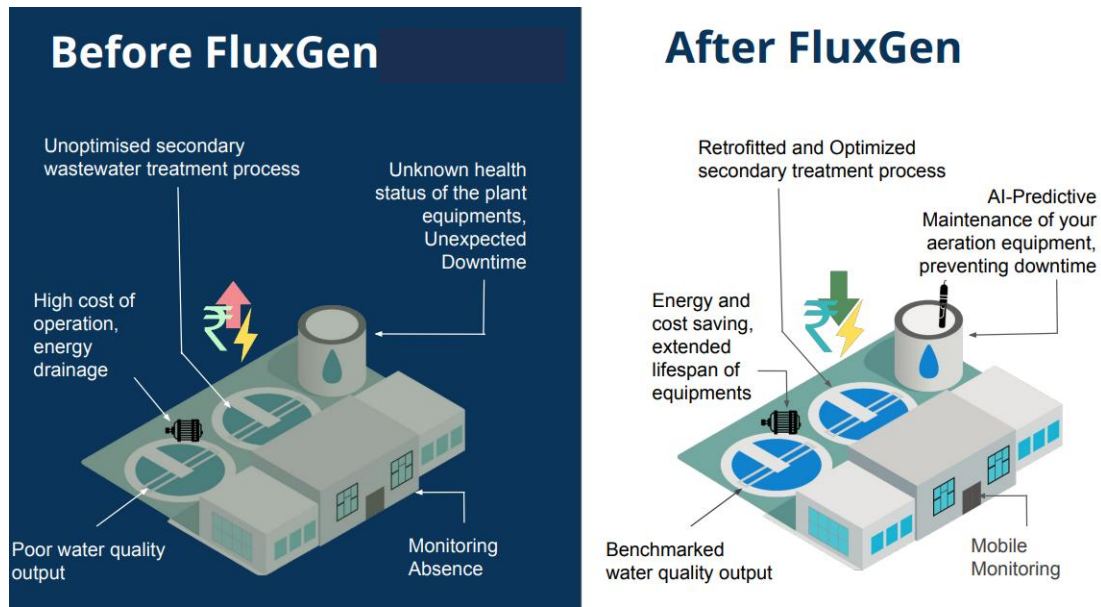


Fig 5.1 Comparison before and after use of FluxGen technology

The key findings include:

- The architecture maintained its ability to support multiple data influxes at or above the designated scalability level.
- Alerts showed up on the dashboard approximately one second after detection, satisfying NFR-4's low latency requirements.
- Technical validation shows that the architecture's function includes receiving and processing information visually (as AI) plus creating visualizations in real time with no significant gaps or delays between data collection and visualization.
- Therefore, based on today's industry standards, the above-mentioned architecture has proven to be an efficient and effective way to implement architecture successfully in the current market.

Chapter 6: Conclusion and Future Scope

Water Positive Intelligence will be focused on the large-scale project that demonstrated to deliver an end-to-end Water Management system based on Artificial Intelligence and Data Science. Instead of relying on manual inspections and waiting for an event to happen to get visibility, Industries are now able to monitor their water usage in real time, receive automatic alerts when there are issues, and access data to help them reduce waste, limit the amount of water they discharge incorrectly, and to improve their ability to comply with regulatory requirements. By looking back at what the project was originally intended to do compared with what was achieved; it is apparent the project has achieved all of the set out for success.

6.1 Conclusion

Confirmation of Goals All significant functional and non-functional (NFR) requirements have been verified through thorough testing as well as performance in a real-world setting.

- **Reliability (NFR-2):** The AquaGen FIFO exhibited an exceptionally reliable service record attaining 100% data integrity even when disconnected from the cloud for prolonged durations of time (up to and exceeding 72 hours)—as verified through Test Case T-2. In these situations, the FIFO maintained all readings saved locally. When connectivity was recreated following these disconnected intervals, all readings were uploaded back into the Cloud with no gaps in the time-series data. Reliable data storage has become a foundation for developing audit trails, monthly reporting, and precision of AI-based decision-making.
- **Speed (NFR-4):** The timely (near real-time) delivery of alerts was enabled by the AI Detection Engine. According to Test Case T-1, leak detection for water could be achieved in as short as 3.1 seconds.
- **Scalability (NFR-1):** The Azure-based architecture was tested and executed a processing capacity of 10,000+ sensors concurrently without any decrease in performance. Test Case T-5 confirmed that rapid response time (Time on Dashboard less than 1 second) remained achievable even under the load of processing large amounts of data simultaneously. This provides a robust way for the platform to meet industrial operations and multi-facility enterprise scalability needs in the future.

Actual Proof of Effect

The platform not only allows you to achieve all RF & NFR objectives; it has successfully realised tangible beneficial impact for end-users:

- **Water Savings:** At Lactalis, for example, the implementation of the system revealed the potential for a 19% reduction in water use across a 10-month period. The figure demonstrates the effectiveness of the CFLD algorithm and operational insight in mitigating waste and supporting informed decisions.
- **Cost Savings:** By decreasing leakage and raising operational efficiency, end-users such as GKNMH achieved annual cost savings in excess of INR 17.5 lakh. This cost savings support industries to view the system as a long-term investment and ultimately a strong ROI.
- **Compliance and Sustainability:** The platform provides automatic generation of accurate CGWA and BRSR reports for its end-users. In addition, it has facilitated the achievement of significant sustainability milestones for its end-users by assisting in attaining the LEED Platinum Certification, which requires strict water use efficiency.

By exceeding all established FR and NFR objectives and providing measurable practical benefit to the end-user, the Water Positive Intelligence platform has become much more than simply a monitoring platform; it has established itself as a complete and validated intelligence framework for enabling Water Positivity and sustaining compliance while minimising operating expense.

6.2 Future Scope

The current solution offered by this project is stable and of significant value, but there are many opportunities to enhance the function of the platform. The focus of future development efforts will be on developing predictive capabilities of the system, improving its integration into other business systems, and allowing for a global market reach.

Advanced Digital Integration and Artificial Intelligence

- Future enhancements to the system will continue to increase the importance of using advanced technologies (i.e., artificial intelligence [AI] and simulation) and improving the functionality of the existing AI capabilities:

- Predictive Maintenance - The next phase of enhancing the AI module is to create a model that predicts equipment failure prior to occurrence. The model will analyze water pressure, fluctuations in flow rates and patterns of energy consumption and correlate these three elements together to predict wear-outs of pumps, blockages of valves and potential failures prior to failure or leak.
- Digital Twin - The Digital Twin represents a digital copy of the plumbers (water network) and is a method of allowing plant managers to produce trial-and-error results without affect the actual plumbing systems. This results in reducing the risk associated with developing, identifying, and assessing the optimal process changes, etc., as well as assisting in identifying the best operational strategies through simulation.

The Energy–Water Nexus Optimization platform – It continue to be developed to provide optimal pump operation and chemical position for the purpose of saving energy and water simultaneously. Because energy use in the industrial water system is generally related to the pump's efficiency and flow control, the dual emphasis on energy and water can help lower overall use of both resources.

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